Transparency without pricing: a credit-level forensic account of collapse in the first tokenized carbon pool

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hen assets of different quality are pooled and sold at a single price, theory predicts that low-quality units drive out high-quality ones, classically only when buyers cannot tell the difference. We document a collapse that occurred despite quality being fully and publicly observable, in a market whose design did not let price reflect it. We test this in the largest pooled carbon-credit market (22 million tonnes of CO_2), whose decline was widely blamed on low-quality forest credits. Analysing every deposit, withdrawal, and account on its public ledger, we find the opposite: 69% renewable-energy credits with little additional climate benefit, and only 4% forest credits. A few accounts extracted most of the high-value credits, while 9.6 million tonnes of low-value credits are now permanently stranded. Identical credits placed in both a screened and an unscreened pool met sharply different fates, consistent with pool design, not credit type, shaping outcomes. Transparency alone therefore appears insufficient to prevent quality collapse: pooled markets may need to price quality, not merely disclose it, a caution for carbon policy and for the $10-billion market now tokenizing other real-world assets on the same design.

Introduction

Every year, companies and governments purchase carbon credits worth billions of dollars to offset their greenhouse gas emissions. A carbon credit represents one tonne of CO_2 prevented from entering the atmosphere, but the climate value of individual credits varies enormously. The central question for any carbon market is whether it can distinguish credits that represent genuine emission reductions from those that do not.

When market design fails to price quality differences, the result is adverse selection: a dynamic in which low-quality goods crowd out high-quality ones, first formalized by Akerlof in 1970 as the "lemons problem"[1,2,3]. The voluntary carbon market (VCM), valued at over $2 billion annually, faces exactly this risk. Fewer than 16% of credits from investigated projects represent genuine reductions[4]. At least 52% of Indian wind power offsets went to projects that would have been built regardless of carbon revenue[5].

Blockchain-based tokenization promised to solve this problem through radical transparency. By recording every transaction on a public ledger, tokenized carbon pools would let any participant verify credit quality before trading. The first and largest such experiment was the Base Carbon Tonne (BCT) pool, operated by Toucan Protocol, which accepted verified credits from the Verra registry (the world's largest carbon-credit standard). Its key design feature: every accepted credit became an interchangeable token priced uniformly, regardless of underlying quality. The pool's value collapsed from approximately $6 per tonne in late 2021 to below $0.50 by mid-2023.

Industry commentary and academic analyses attributed this collapse to low-quality forest conservation credits, specifically REDD+ (a program that pays landowners to avoid deforestation)[2,6,7,8,9]. This narrative drew support from the broader literature documenting REDD+ crediting problems[4,10,11]. But it was never tested against the pool's actual composition data, which are publicly recorded on the blockchain and straightforward to query. More broadly, no study has empirically tested the dominant policy assumption that transparency prevents quality collapse.

Here, we analyze every deposit (1,187), every withdrawal (35,432), every credit token (161 individually scored, of 345 total), and every market participant (509 depositors, 28,897 redeemers) across BCT's entire operating history. The data contradict the prevailing narrative. BCT contained 69.1% renewable energy credits (predominantly grid-connected hydropower, with wind and solar, registered across China, India, Turkey, and Brazil) with near-zero additionality (meaning the projects would have been built with or without carbon revenue), and just 4.2% REDD+. A small number of accounts extracted most of the high-demand credits (1.55 million tonnes), while 9.6 million tonnes of low-quality renewables remain permanently stranded.

A comparison of identical credits across two pools with different quality screens sharpens the finding. The same credits were 100% redeemed from BCT but only 28.5% from a quality-gated pool, a contrast consistent with pool design, rather than credit type, shaping asset fate. Unlike prior work that quantifies overcrediting at the project level[4,5], we trace how overcredited assets are selectively extracted once they enter a pooled market, even when quality differences are publicly observable.

We term this outcome *design-enabled adverse selection*. It is distinct from Akerlof's classical lemons problem, which requires private information about quality. Here, quality information is available to all participants but is not reflected in prices. The pool treats every deposited credit as identical: a participant cannot buy a specific high-quality credit, only a generic pool unit priced at the pool average. We argue that uniform pricing of heterogeneous assets under voluntary participation is sufficient to produce market failure.

Because every transaction in this market is publicly recorded, its complete ledger supports the first credit-level forensic account of this dynamic, the kind of reconstruction that opaque off-chain carbon markets cannot supply. We treat the market as a case study of a general prediction, not a system of intrinsic interest. The implication for pooled markets broadly, including the $10 billion tokenized real-world asset sector and the Article 6.4 crediting mechanism (a new international carbon trading system being negotiated under the Paris Agreement), is that transparency without price differentiation appears insufficient for market function.

Results

BCT's real composition contradicts the REDD+ narrative

We extracted all 1,187 deposit records from BCT's public transaction ledger, covering 168 unique Verra VCS (Verified Carbon Standard) projects and approximately 22 million tonnes of tokenized carbon credits (see Methods).

Renewable energy credits dominated the pool (Fig. 1). Grid-connected hydro, wind, and solar projects (predominantly from China and India, with Turkey and Brazil) registered between 2008 and 2013, constituted 69.1% of total deposited tonnage. These are precisely the categories from the CDM era (the Clean Development Mechanism, a now-defunct UN crediting program, 2006–2020) that the ICVCM (Integrity Council for the Voluntary Carbon Market, the sector's main governance body) has declined to approve for its Core Carbon Principles (CCP) label. Multiple independent studies document their near-zero additionality: Cames et al.[12] and Schneider et al.[13] in program-level assessments, Probst et al.[4] in a meta-analysis finding no statistically significant emission reductions from wind power CDM projects, and Calel et al.[5] estimating 52% non-additionality among Indian wind offsets. The projects would have been built regardless of carbon revenue.

As an illustrative, resolution-limited cross-check, a satellite methane analysis of 9 BCT waste/methane projects found 5 of 9 with concentrations not significantly below background; we treat this only as suggestive and do not rely on it for the composition findings (Supplementary Methods).

REDD+ credits, widely blamed for BCT's collapse, constituted just 4.2% of the pool, more than sixteen times smaller than the renewable share. Even combining all nature-based categories (REDD+, ARR (afforestation, reforestation, and revegetation), and IFM (improved forest management)), the total reaches only 10%, versus 69.1% for renewables alone.

Base-rate selection: BCT over-selected renewables at 1.87× the registry rate

BCT's renewable dominance could reflect the broader carbon credit supply rather than a selection effect. We obtained Verra VCS issuance data from MSCI (2024) and Ecosystem Marketplace (2023) indicating that renewables constitute approximately 37% of total VCS issuance by volume (central estimate; robustness tested across the full sensitivity range of 26–48%).

BCT's renewable share (69.1% by tonnage) is 1.87× the VCS base rate (selection coefficient: the ratio of BCT's renewable share to the base rate; 1 = no over-selection). Because deposits cluster by account, we use account-clustered inference rather than a naive binomial test: an account-level permutation test (10,000 iterations) produces $p < 0.0001$, and 89.0% of accounts hold majority-renewable portfolios. The over-selection is robust to conservative base-rate assumptions (remaining 1.43× at a 55% null) and to account-level bootstrap and HHI-adjusted tests (Supplementary Methods).

Conversely, REDD+ is under-represented at 0.2× the VCS base rate (BCT 4.2% vs. VCS ∼17%), directly contradicting the narrative that REDD+ credits overwhelmed the pool.

Bridge-level decomposition. Toucan's bridge (the mechanism that moves an off-chain credit on-chain) passed essentially every Verra-bridged credit into BCT, 93.5% of tokens and 99.6% of tonnage (Supplementary Methods), so the 1.87× over-selection was a bridge-level phenomenon: the permissionless (open to anyone) bridge disproportionately attracted renewable credits from Verra, rather than depositors selecting them from a diverse universe. Pool-level selection space was essentially zero.

Price-quality feedback loop: composition predicts price collapse

We next ask whether quality degradation drove the price collapse. We merged daily BCT prices in US-dollar terms (828 observations, October 2021 – July 2024; see Methods) with daily cumulative quality metrics from the deposit stream ($n = 331$ overlapping days).

Correlation. BCT's price and cumulative Pool Quality Deficit (PQD, a tonnage-weighted 0–1 measure of how far a pool's credits fall below maximum quality; Methods) are strongly correlated (Pearson $r = 0.774$, $p < 10^{-66}$). In a first-differenced OLS regression (analyzing day-to-day changes rather than levels, the credible specification for non-stationary series), each percentage-point increase in the renewable share of deposits predicts a \$1.80 decrease in BCT's price ($\beta = -1.8$, $p < 0.001$).

Redemption-side evidence. Analysis of 35,432 Transfer events from the BCT pool contract reveals that selection operated on both sides. Non-renewable credits were preferentially redeemed (Table 1): industrial gas 100%, REDD+ 99.8%, IFM 93.0%, ARR 91.3%, versus just 3.7% of renewable credits. The tonnage-weighted mean composite quality score (0–100, higher = better) of redeemed credits (38.7) exceeded that of deposited credits (31.7) by 7 points, a 22% premium. As a result, BCT's net renewable share increased from 71% to 76% over the pool's lifetime.

The exit followed a type-level Gresham sequence (bad drives out good; Fig. 2b): the most valuable credit types were extracted first, in an order matching off-chain market demand ($\rho = -0.74$, $n = 7$ types; Supplementary Methods). Before the May 2022 crash, redemptions targeted high-value credits (tonnage-weighted quality 42.0); afterwards quality dropped to 32.4, reflecting selective exit while BCT's price still exceeded the off-chain value of premium credits.

Depositors and redeemers are almost entirely distinct populations: only 1.4% of 28,897 redeemer accounts also appear among the 509 depositor accounts. The top 10 redeemers account for 85.0% of extracted tonnage. Their transactions occurred in rapid automated bursts (median gap 4.6 seconds), consistent with programmatic execution.

This dual-margin structure — indiscriminate, architecture-driven entry (99.6% tonnage pass-through) on one side; deliberate, type-specialist extraction on the other — is not predicted by standard adverse-selection models, which assume a single population.

Account forensics: who extracted and what they took

Account-level analysis of the 161 scored tokens' transfer histories reveals that the largest redeemers are type-specialist extractors who never participated in the deposit side.

Table 2. Top 5 redeemers by tonnage.

Account	Tonnes redeemed	Dominant type	Mean quality	Also depositor?
0x65a5...	651,334	Industrial gas	30.9	No
0x1b8e...	335,556	IFM	40.0	Yes
0xee4b...	195,051	REDD+	31.4	No
0x4b3e...	188,205	ARR	50.4	No
0x8556...	183,629	ARR	46.0	No

These five accounts removed 1.55 million tonnes: the vast majority of redeemed industrial gas, 37% of REDD+, and 44% of ARR. The largest by tonnage (0x65a5...; 651,334 tonnes of industrial gas, 42% of the 1.55 million) is not an arbitrageur but a smart contract (verified on-chain; see Data availability) that retired every credit in the same transaction as redemption, the signature of a retirement aggregator routing credits to final use. The other four are individual wallets whose redeem-then-hold behaviour is consistent with strategic arbitrage of the gap between BCT's uniform price and higher off-chain values, an estimated \$4–\$8 million in captured differential (an upper bound, as it includes retirement-routed volume; see Methods). Fifteen of the twenty largest redeemers never deposited anything, and each targeted a single credit type.

Tracing each redeemed credit's immediate destination across the 20 largest extractors shows three pathways: cross-pool transfer to a quality-screened pool (39.2%), immediate retirement (34.6%), and secondary-market sale (17.0%). The cross-pool movement followed a strict, systematic ordering (deposit, redeem, then re-deposit into the screened pool; Supplementary Methods).

Among the 399 accounts that both deposited and redeemed (the 1.4% overlap population), a quality-swap analysis reveals no systematic quality arbitrage (Supplementary Methods). The dual-margin mechanism thus operates through largely distinct account populations connected by the pool's uniform pricing. A first-funder audit of the top 20 accounts per margin found entity-level links for a minority: one shared funding source, direct credit transfers among three accounts, and two dual-margin accounts, together 26% of top-20 deposit and 9% of top-20 redemption tonnage (Supplementary Methods). The separation is therefore account-level, not entity-level.

Quality atlas and quality gating

BCT's PQD of 0.679 places it at the bottom of the VCM quality spectrum, which spans a 10-fold range from 0.076 (DACCS, direct air carbon capture and storage) to

0.759 (grid-connected renewables) (Fig. 3). A within-pool permutation test ($z = -0.64$, $p = 0.27$) finds no evidence of within-pool selection bias; the eligible universe was uniformly low-quality. A BBB quality gate (letter grades B–AAA map the composite score to bands, bond-rating style; the BBB threshold separates credits with verified additionality from those without) would have substantially reduced this deficit, by about a third, from 0.724 to 0.405, in a counterfactual simulation (Fig. 6). The Toucan CHAR pool (biochar allowlist, PQD 0.221) demonstrates that category restriction prevents quality collapse (Supplementary Fig. 5).

Temporal dynamics and robustness

Deposit quality declines over BCT's operating life (Spearman $\rho = -0.24$, $n = 1{,}187$), but this is entirely a vintage-composition effect: late deposits drew from progressively older vintages (vintage, the credit's emission-reduction year), not from intrinsically worse credits within vintage (Supplementary Table 2). The May 2022 cryptocurrency crash provides exogenous deposit-side evidence of the causal channel: post-shock deposit quality fell 3.4 points (Cohen's $d = 0.49$, a moderate effect) as volume collapsed by 98% and the quality-decline slope tripled, consistent with the price-to-quality feedback above (Supplementary Methods).

Pool design and asset fate: a within-token cross-pool contrast

A sharper contrast on the same data comes from 14 credit tokens (1.50 million tonnes) deposited into *both* the unscreened pool (BCT) and the quality-screened pool (NCT), so the same physical credit appears under both designs with quality, vintage, registry, and project type held fixed. These 14 are the redeemed members of the 31 credits shared by the two pools, so the unscreened-side rate is near-universal (100%) partly by this selection, while the screened-side rate (28.5%) is not symmetrically conditioned; the +73.9-percentage-point gap below is therefore best read as an upper bound on the contrast. Token by token (the unit of inference, $n = 14$), the mean redemption gap is +73.9 percentage points (95% CI [51, 93]; Fig. 4), unanimous in direction across every token and every credit type, with three exact tests (sign, permutation, Wilcoxon) returning $p = 2.4 \times 10^{-4}$. This is a two-pool contrast, not 14 independent treatments, and shares the two-pool confounds (exposure timing, liquidity, eligibility) that make the pool-level analysis descriptive only (Methods); its scope is also narrow, covering only nature-based credits. We therefore read it as *strongly suggestive* of a pool-design effect rather than a proof, though robust: a hidden confounder would have to more than triple the odds of redemption for one credit over its pair to overturn the result (Supplementary Methods).

Cross-pool comparison. Recovering per-token credit identity on-chain for five pools across two operators (Toucan and C3, a second bridge protocol) and scoring each by our common framework, mean deposit quality rises monotonically with screening strictness, from the unscreened pools (∼29–31) through nature-screened pools (∼39–40) to the category-allowlist pool (77.9; Supplementary Table 4). This gradient is, however, in large part definitional: a screen is defined on credit type and our composite is itself type-driven, so screened pools must score higher; no within-type comparison holds type fixed, and the quantity is entirely framework-derived (the C3 pools have no external validation). The unscreened pools also show a declining temporal slope absent in screened pools, but these raw slopes are not vintage-controlled and may reflect the same vintage drift we identify for BCT above. We therefore read the cross-pool pattern as corroborative and descriptive only, not as evidence that design causes quality.

Market-revealed quality hierarchy

Credit type alone, not the quality framework, captures the dominant axis of selective redemption: a framework-free type-only prediction rule out-predicts the quality-grade rule, and within the 116 renewable tokens neither vintage nor grade predicts redemption (Supplementary Table 3). The paper's core findings (composition, base-rate over-selection, redemption-side extraction, and the within-token cross-pool comparison) accordingly rest on transaction data and Verra credit-type classifications, not on the quality scoring framework.

In sum, 9.6 million tonnes of B-grade credits (63% of the 15.2 million tonnes covered by the 161-token scored analysis) remain unredeemed, one of the largest carbon credit graveyards in existence.

Discussion

Transparency without price differentiation does not prevent quality collapse. Every transaction was public and every credit's quality metadata freely available on the Verra registry and the blockchain. Yet the market collapsed exactly as economic theory predicts for markets with *hidden* quality differences.

The underlying principle is Gresham's Law ("bad money drives out good"), which Akerlof (1970)[1] generalized as the "lemons problem" for markets where quality is unobservable. BCT shows the same dynamic operates even when quality *is* observable: the mechanism we term design-enabled adverse selection runs through market architecture rather than information asymmetry, identifying a distinct pathway to lemons-like outcomes. Our within-token contrast provides credit-level descriptive evidence consistent with the lemons dynamics that Manshadi et al.[2] model theoretically.

Stated generally: whenever heterogeneous assets are pooled at a single clearing price, with voluntary entry and exit, and the highest-quality units retain higher value outside the pool, those units are selectively withdrawn while low-quality units accumulate, *independently of whether quality is observable.* Observability changes who profits from the spread, not whether the spread exists.

The misattribution of BCT's collapse to REDD+ credits[7,8,9] illustrates how structural quality failures can hide in plain sight. CDM-era renewable energy credits constituted 69.1% of BCT's tonnage. They suffered from near-zero additionality because the underlying projects were already economically competitive. Unlike REDD+ failures, which generate visible controversy over baselines (the hypothetical emission levels that would have occurred without the carbon credit project) and deforestation, the additionality failure of renewable energy credits is structurally invisible: the dams and wind farms exist, the electricity is real. Quality assurance frameworks that focus on high-profile failure modes may miss the categories that dominate pool composition.

Nearly 10 million tonnes of nominally valid offsets remain permanently stranded at a 97.6% non-redemption rate (within the 161-token scored subset), an illustrative welfare gap of \$100–\$250 million (Monte Carlo median \$146M, 90% CI \$68M–\$252M). Meanwhile, a small number of type-specialist accounts captured an estimated \$4–\$8 million (an upper bound; see Results) by selectively redeeming the credits that off-chain markets still valued, though the single largest such address is a retirement contract, not a profit-taking arbitrageur. Where Probst et al.[4] and Calel et al.[5] quantify overcrediting at the project level, we show how it translates into participant-level extraction inside a pooled market. The gap measures a compositional shortfall against a representative counterfactual pool; the stranding effect concerns circulation, so the two frames are complementary rather than contradictory. Paradoxically, BCT's failure functioned as an inadvertent quality filter, removing 9.6 million tonnes of low-quality credits from active circulation, a permanent illiquidity event that no registry-level intervention has achieved.

The underlying logic (Gresham's Law under uniform pricing) applies wherever heterogeneous assets are pooled[3,14]; the same dual-margin pattern appears, observationally, in two other uniform-priced pooled markets we examined (a crypto-asset pool and an NFT pool; Supplementary Methods).

These findings apply most directly to voluntary pooling and registry-based bundling of credits, where quality-heterogeneous units clear at a common price[14,15]. The Article 6.4 mechanism under the Paris Agreement is a sovereign, quantity-fixed system rather than a supply-driven pool, so it inherits the design principle (price quality, or expect collapse) rather than a direct prediction. Recovery from a low-type equilibrium requires discontinuous intervention: even a minimal quality floor (composite score $\geq$29) would have excluded the worst half of BCT's tonnage. These results point to three design principles for future pooled crediting mechanisms: quality-differentiated pricing tiers rather than single-price pools; dual-margin gating of both deposits and withdrawals (exit-side extraction proved at least as damaging as entry-side quality loading); and dynamic quality floors that adjust as composition shifts. Our findings provide direct empirical support for the ICVCM's Core Carbon Principles framework[16], and the emerging quality-differentiation ecosystem (Singapore's ratings mandate[17], blockchain credibility indices[18], quality-gating protocols[19]) must incorporate dual-margin design to be effective.

Beyond diagnosis, the same public data support a deployable early-warning. A cumulative Lemons Index (Lemons Index, 0–1, higher = worse average pool quality) computed in real time from the deposit stream (each day using only deposits already observed) crossed a 0.50 danger threshold at the pool's launch (already at 0.71 while the price was still near its launch level of \$5.91), roughly nine months before the price more than halved. Because this is a standing *level* signal, it does not depend on price history: the diagnostic information was in the composition from inception (exploratory temporal-ordering tests are reported in Supplementary Methods). Paired with the quality gate (in a counterfactual simulation, a BBB floor cuts the pool's quality deficit by about a third, from 0.724 to 0.405; Fig. 6), this converts the post-mortem into an actionable protocol: audit composition at inception, gate deposits and redemptions on quality, and re-audit as the pool evolves.

Several limitations warrant acknowledgment. Our quality scores are author-derived, but the core findings rely on transaction data and Verra credit-type classifications rather than on the scoring framework, which is independently validated against CCP labels (d = 1.8, a very large separation; Fig. 5), commercial ratings (ρ = +0.901, n = 27; Supplementary Fig. 1), and a large-language-model panel (Fleiss' κ = 0.6, moderate agreement; Table 4). Both cross-pool comparisons are descriptive, not causal identifications: the within-token contrast (+73.9 pp gap, Γ = 3.05 bound) and the pool-level difference-in-differences rest on only two pools, so their statistics quantify per-pair variability rather than the pool-level confounds (exposure timing, liquidity, eligibility) that remain collinear with the screening label (Methods). We read the contrast as strongly suggestive of a pool-design effect, noting that the dominant renewables are structurally excluded from this nature-based comparison. The bridge decomposition (99.6% pass-through by tonnage) establishes that entry-side selection was primarily architectural rather than strategic.

This result carries a falsifiable policy implication: any pooled crediting system providing quality transparency without quality-differentiated pricing will experience the same collapse. The transparency reforms by ICVCM and Article 6.4 negotiators are necessary but not sufficient: if Article 6.4 pools heterogeneous credits at a single price, the same Gresham dynamic should emerge, and mechanisms with quality-differentiated tiers should not. This prediction is testable as Article 6.4 becomes operational.

The tokenized real-world asset sector now exceeds $10 billion and is replicating pool-based architecture across commodities, bonds, and receivables. BCT is the first pool to complete a full quality-collapse cycle with auditable transaction data. Its lesson is straightforward: quality-differentiated mechanism design is not an optional refinement for pooled carbon markets but a structural prerequisite for market integrity.

Methods

We collected all deposit transactions to the Toucan Protocol BCT pool contract on the Polygon blockchain using a Polygon RPC endpoint. We identified 1,187 deposit transactions spanning the period from BCT's launch in October 2021 to the effective cessation of new deposits in late 2022, mapping to 168 unique VCS project identifiers (345 unique TCO2 token addresses) and approximately 22 million tonnes of tokenized carbon credits. For each deposit, we recorded: transaction hash, block number and timestamp, depositor address, TCO2 token contract address, Verra VCS project identifier, vintage year, and tonnage. Project identifiers were cross-referenced against the Verra registry to obtain methodology category, country of origin, and CCP eligibility status.

Project classification. Each project was classified into one of eleven methodology categories based on the Verra registry entry: renewable energy (grid-connected wind, hydro, solar; CDM methodologies AMS-I.D, ACM0002), fossil fuel switching, waste management and methane capture, REDD+ (VM0007, VM0009, VM0015), afforestation/reforestation (ARR), industrial gas destruction, improved forest management (IFM), energy efficiency, industrial processes, agriculture, and cookstoves. Pool-level composition statistics were computed with tonnage weighting.

NCT data. We collected all 708 deposit events from the Toucan NCT pool on the same Polygon blockchain over the same block range. NCT restricts deposits to AFOLU (agriculture, forestry, and other land use) credits with vintage ≥2012, functioning as a minimal quality gate on project type. NCT and BCT share identical protocol infrastructure and blockchain substrate.

Price data. Daily BCT prices denominated in USDC (a stablecoin pegged to the US dollar) were obtained from the DeFi Llama API for the Polygon SushiSwap BCT-USDC pair (828 observations, October 2021 – July 2024).

Quality scoring framework

We developed a composite quality scoring framework evaluating six active dimensions (chosen to cover the primary determinants of credit integrity formalized by the Oxford Offsetting Principles and the ICVCM CCP Assessment Framework, the two most widely cited normative hierarchies for credit quality), each scored on a 0–100 scale: removal type hierarchy (weight 0.250), additionality (0.200), MRV (monitoring, reporting, and verification) grade (0.200), permanence (0.175), vintage year (0.100), and registry and methodology quality (0.075). Initial weights were derived from the Oxford Offsetting Principles[20] and the ICVCM CCP Assessment Framework[16]. A pilot Best-Worst Scaling expert panel ($n = 5$) confirmed consistency with our weights (Spearman $\rho = +0.814$). Monte Carlo perturbation (10,000 Dirichlet-sampled weight vectors) confirms 93.7% grade stability. The composite was computed as $C = \sum_i w_i \times s_i$ and mapped to letter grades: AAA (≥90), AA (≥75), A (≥60), BBB (≥45), BB (≥30), B (¡30). Seven disqualifier conditions cap maximum grades regardless of composite score. The co-benefits dimension was assigned zero weight to avoid amplifying adverse selection through co-benefit narratives[6]. The Pool Quality Deficit ($\mathrm{PQD} = 1 - \bar{C}/100$) quantifies tonnage-weighted quality degradation per pool. Full scoring details, distributional scoring model, extended scoring procedures, and sensitivity analyses are provided in Supplementary Methods.

Base-rate comparison

BCT's renewable energy share was compared against the Verra VCS base rate (central estimate 37%, sensitivity range 26–48%, from MSCI[21] and Ecosystem Marketplace reports). We define the selection coefficient as the ratio $S = f_{\mathrm{BCT,renewable}}/f_{\mathrm{VCS,renewable}}$ (a quantity computed for this purpose, not a borrowed named statistic), tested via exact binomial test, with account-clustered robustness (permutation, HHI-adjusted, and bootstrap tests; see Supplementary Methods).

Within-token matched-pair design (descriptive cross-pool contrast)

The within-token pool-design contrast is computed from the 14 TCO2 tokens deposited into *both* BCT (unscreened) and NCT (quality-screened). Because the same physical credit appears in both pools, quality, vintage, registry, and project type are held fixed by construction and the only difference within a pair is pool architecture; each credit is its own control. Per-token deposit and redemption tonnage for both pools were reconstructed directly from raw ERC-20 Transfer logs (deposit = wallet→pool; redemption = pool→wallet) cached per token, reproducing the published aggregate rates (100.0% / 28.5%) to within 0.01 percentage points and confirming the NCT per-token rates are measured, not imputed. Treating the token as the unit of analysis ($n = 14$), we tested the per-token redemption-rate difference (BCT − NCT) with three exact procedures (a paired sign test, a full-enumeration permutation test over per-token sign flips, and a Wilcoxon signed-rank test) and quantified the cross-token mean gap with a percentile bootstrap over the 14 per-token differences (20,000 resamples). The exact-test p-value reflects the unanimous direction of the gap and is unchanged whether the one near-tied token (BCT redemption within ≈1 tonne of 100%) is dropped or retained. A tonne-level Beta–Binomial model (deposited tonnes as Binomial trials, uniform Beta(1,1) priors) was also fit, but it assumes independence across tonnes whereas redemptions occur in lumpy batches; it understates uncertainty and is reported only descriptively, with inference based on the token-level bootstrap and exact tests. The estimand is the pool-design effect on dual-eligible credits (those admitted by both pools), a higher-quality nature-based subset rather than the full unscreened pool. Robustness to unobserved confounding was quantified with a Rosenbaum-style sensitivity bound (the odds-ratio Γ at which the sign-test result loses significance). A mixed-effects logit with a token random intercept was also fit but exhibits quasi-complete separation (BCT ≈ 100% redemption) and is reported only as directional corroboration. The selection channel (which credits reached both pools) is bracketed by the sensitivity bound and a directed acyclic graph (Supplementary Fig. 3).

Pool-level difference-in-differences (descriptive only)

A deposit-level DiD pooling all 1,895 events (1,187 BCT + 708 NCT) was also estimated:

$$\text{quality}_i = \beta_0 + \beta_1 \cdot \text{BCT}_i + \beta_2 \cdot \text{PostShock}_i + \beta_3 \cdot (\text{BCT} \times \text{PostShock})_i + \varepsilon_i$$

where β_3 captures whether BCT's quality changed *differently* from NCT's after the May 2022 crash, with standard errors clustered at the TCO2 token level. We report it as descriptive corroboration only, not as a causal estimate: BCT and NCT share 88.6% of their tokens, violating the unit-independence its inference assumes. Under a token-clustered bootstrap the standard error inflates 4.02-fold and the between-pool rank difference is indistinguishable from zero (cluster-robust $p = 0.24$, versus naive permutation $p < 0.0001$). Causal identification therefore rests on the within-token design above.

Multi-pool generalization

To assess whether the mechanism generalises beyond a single pool, deposits to two further bridge instruments (C3 and Moss MCO2) were scored by the same methodology-archetype procedure used for BCT. For BCT and NCT we report the temporal-quality slope (Spearman ρ of composite vs deposit order) from the within-operator permissionless-versus-filtered contrast ($\rho = -0.439$ vs $\rho = -0.105$). For C3, per-token Verra project and vintage are encoded in the token symbol and were recovered on-chain via `eth_call` (22/26 tokens genuine; 17/18 projects typed, five via public-registry lookup; 75/92 deposits scored); C3's temporal-quality slope is $\rho = -0.329$ ($p = 0.004$), matching BCT's direction. Moss MCO2 is a single fungible basket token with no per-token credit identity, so a per-token quality signal is genuinely undefined and adverse selection is structurally inapplicable. Remaining coverage gaps are documented explicitly rather than imputed.

Redemption analysis

We identified redemptions from ERC-20 Transfer events where the BCT pool contract appears as the sender, yielding 35,432 events across the 161 originally scored tokens (15.2M of 22M total tonnes). Redemption rates were computed per credit type as redeemed/deposited tonnage. The within-token cross-pool comparison exploited 14 tokens deposited into both BCT and NCT to test whether the same credits experienced different fates under different pool designs. Account forensics, destination tracing, and profit quantification methods are detailed in Supplementary Methods.

Price-quality dynamics

Cumulative PQD and renewable share were merged with daily BCT prices ($n = 331$ overlapping days). A first-differenced OLS with Newey–West HAC standard errors (10 lags) tested contemporaneous co-movement. Bidirectional Granger causality was tested at weekly frequency ($n = 55$) using VAR(2) selected by AIC. We acknowledge that the small sample limits statistical power; the daily regression ($n = 330$) provides the more robust test.

Cross-domain comparison

To test whether the adverse selection pattern generalizes beyond carbon markets, we analyzed composition shifts in the Curve Finance stETH/ETH liquidity pool on Ethereum during the May 2022 market crash. Curve's stableswap treats stETH (a liquid staking derivative) and ETH as near-equivalent, a uniform-pricing mechanism analogous to BCT's 1:1 tokenization. We computed weekly net flows for each asset and defined the composition shift as the absolute difference in net flows (ETH outflow minus stETH outflow). The pre-crisis baseline was the mean weekly absolute composition shift over the 8 weeks preceding 7 May 2022. This analysis is presented as supporting evidence; the Curve comparison is observational and also consistent with a classical bank-run interpretation.

Welfare gap estimation

We estimated the welfare cost of BCT's quality degradation using a Monte Carlo simulation (10,000 iterations), chosen because multiple uncertain inputs (additionality rates, social cost of carbon, counterfactual composition) must be propagated simultaneously. Each iteration sampled: (i) additionality rates per credit type from literature-derived distributions[4,12] (renewable energy: 0–20%; REDD+: 25–75%; IFM: 30–70%; ARR: 40–80%); (ii) the social cost of carbon from a uniform distribution (\$50–\$200/tCO_2, spanning the range used by the US EPA and the Stern Review); and (iii) a counterfactual pool composition matching VCS registry base rates. The welfare gap was computed as the difference in expected climate value between the counterfactual pool and BCT's actual composition. This estimate is illustrative and conditional on both the additionality assumptions and the counterfactual specification.

Statistical inference

All p-values were corrected using the Benjamini–Hochberg FDR procedure at $\alpha = 0.05$ across the following 10 primary tests: (1) base-rate selection coefficient (binomial), (2) full-sample temporal correlation, (3) pre-shock temporal correlation, (4) within-token matched-pair sign test, (5) pool-level between-pool difference (descriptive, cluster-robust), (6) Granger quality→price, (7) Granger price→quality, (8) within-pool permutation, (9) depositor concentration Mann–Whitney, (10) redemption differential chi-squared. Headline claims were supplemented with 10,000-permutation p-values. Cluster-robust bootstrap (clustered by depositor account, 10,000 iterations) was used for deposit-level statistics. Additional inference details (account-clustered tests, CCP calibration, inter-rater reliability, rank correlations with commercial agencies) are provided in Supplementary Methods.

Data availability

All data, scoring rubrics, analysis scripts, and smart contract source code are available at https://github.com/Adeline117/carbon-neutrality under an MIT licence. Machine-readable rubrics are in JSON format. The 318-credit methodology batch dataset with per-dimension scores is included. All 345/345 BCT tokens are scored (161 original + 184 imputed). On-chain deposit data for both BCT (1,187 transactions) and NCT (708 transactions) are provided with transaction hashes enabling independent verification. The externally-owned-account versus smart-contract status of the top redeemers was checked with `eth_getCode` against a public Polygon node; results are recorded in `redeemer_contract_check.json`. The entity-level independence audit (first-funder resolution for the top 20 accounts per margin) is reproducible via `entity_funding_analysis.py`, with per-wallet results in `entity_funding_analysis.json`. LLM panel outputs for all models across 29 credits are included. Analysis scripts are pure Python with NumPy/SciPy as sole dependencies.

Code availability

All analysis code is available at https://github.com/Adeline117/carbon-neutrality under an MIT licence.

Author contributions

A.W. designed the study, collected and analyzed all data, developed the quality scoring framework, performed all statistical analyses, and wrote the manuscript. W.C. supervised the research and provided critical revisions.

Competing interests

The authors declare no competing interests.

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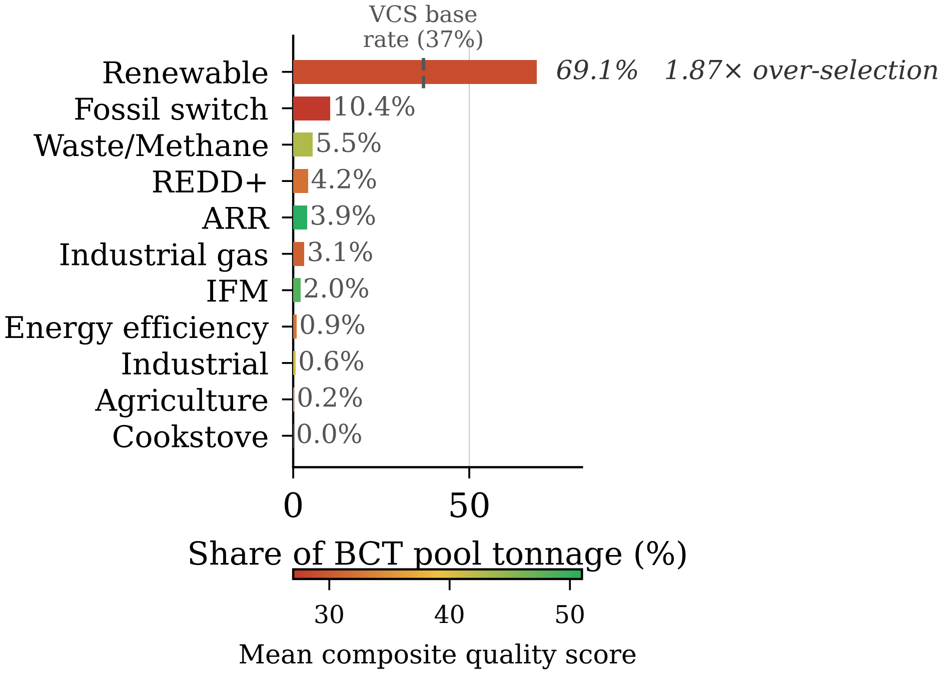

Fig. 1 BCT pool composition by methodology category, weighted by deposited tonnage. Renewable energy credits constitute 69.1% of total deposited tonnage; REDD+ constitutes just 4.2%.

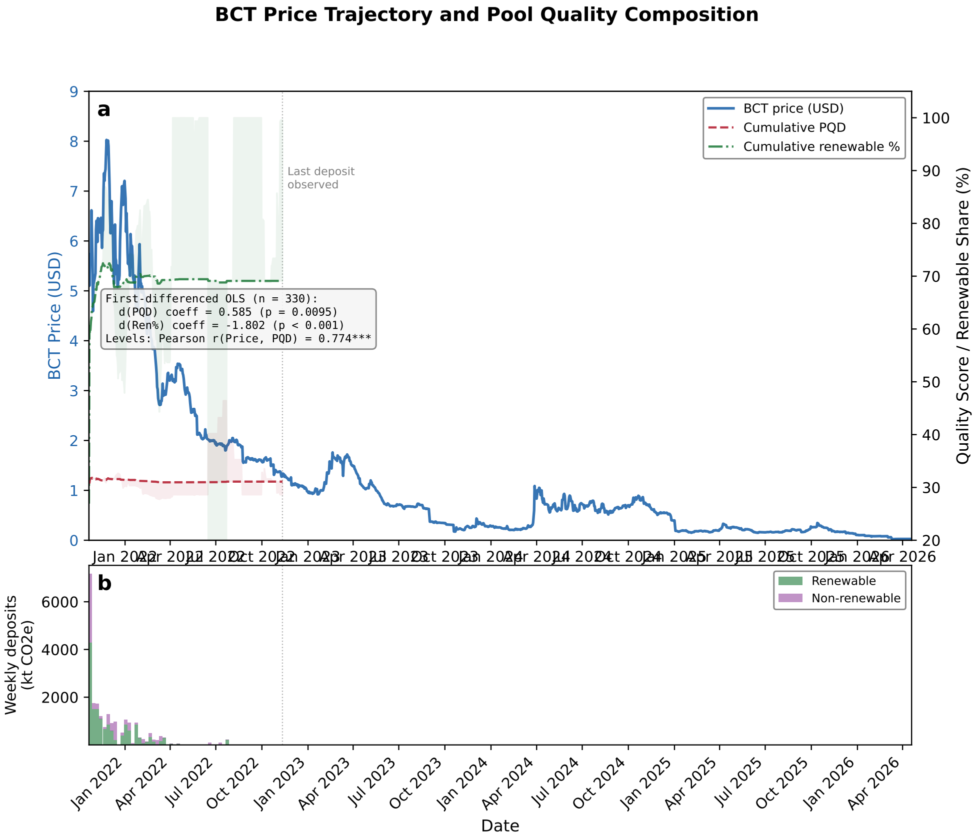

Fig. 2 BCT price trajectory and pool quality composition. (**a**) Daily BCT price against cumulative Pool Quality Deficit and cumulative renewable share; inset reports the first-differenced OLS (n = 330) and the levels correlation (Pearson $r = 0.774$). (**b**) Weekly deposit volume split into renewable and non-renewable tonnage.

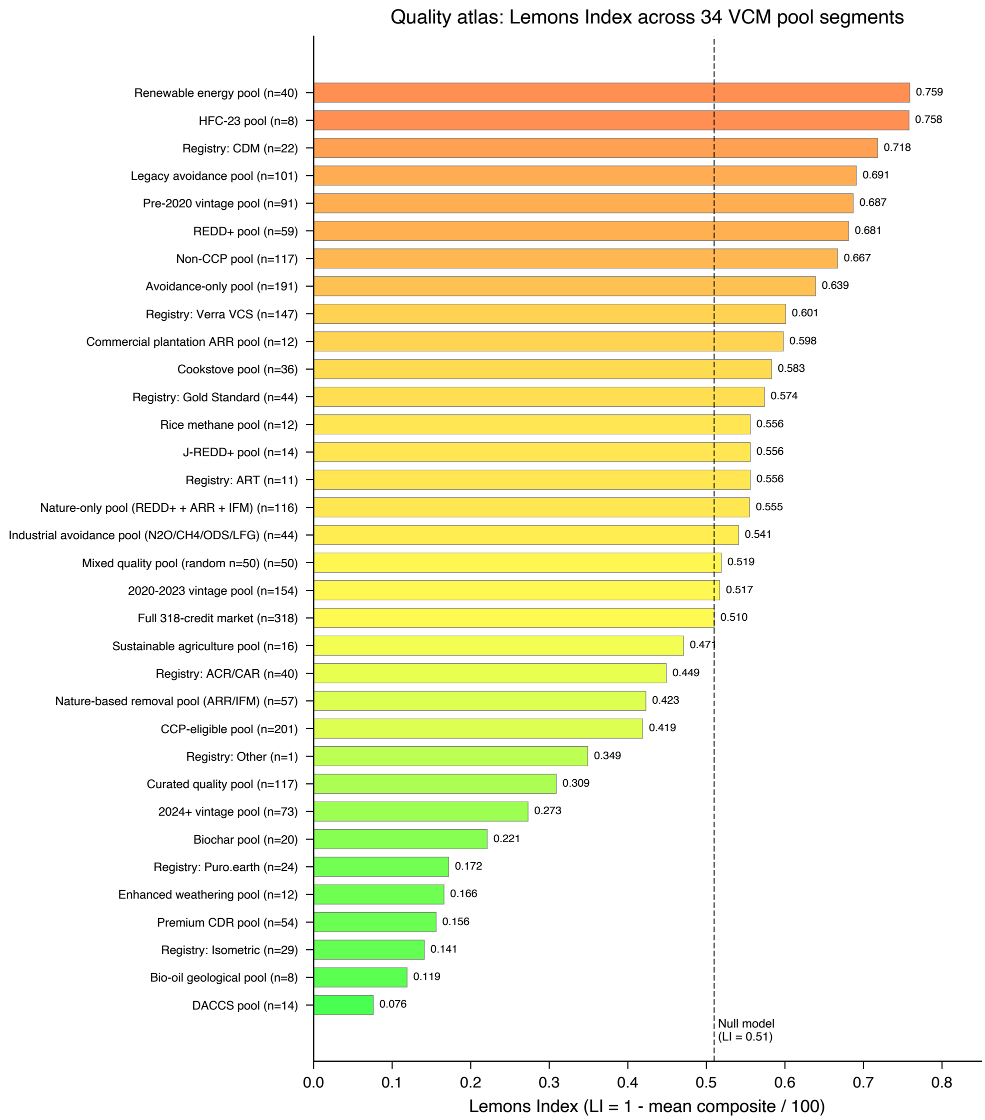

Fig. 3 Quality atlas: 34-segment PQD spectrum with null model baseline. PQD ranges from 0.076 (DACCS) to 0.759 (grid-connected renewables).

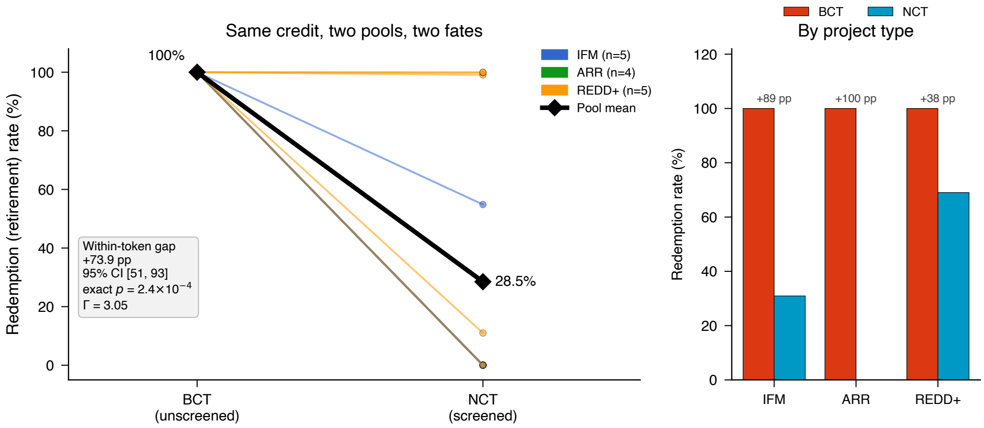

Fig. 4 Within-token cross-pool contrast: redemption of identical credits under two pool designs (descriptive; a two-pool comparison, not a causal identification). (**a**) Per-token slope plot of redemption (retirement) rate for the 14 carbon credit tokens deposited into *both* the unscreened BCT pool and the quality-screened NCT pool, coloured by project type; the bold black line traces the pool means (100.0% redeemed from the unscreened pool versus 28.5% from the screened pool). Across the 14 pairs (the unit of inference) the mean within-token redemption gap is +73.9 percentage points (token-level bootstrap 95% CI [51, 93]); the direction is unanimous, so exact sign, permutation, and Wilcoxon tests all return $p = 2.4 \times 10^{-4}$ whether or not the one near-tied credit is included. (**b**) Redemption rate by project type (tonnage-weighted): the gap is positive within every type (IFM +89, ARR +100, REDD+ +38 pp). The result is robust to unobserved confounding up to a Rosenbaum bound of $\Gamma = 3.05$. Depositor-level temporal, base-rate, and difference-in-differences robustness panels are reported in Supplementary Fig. 3.

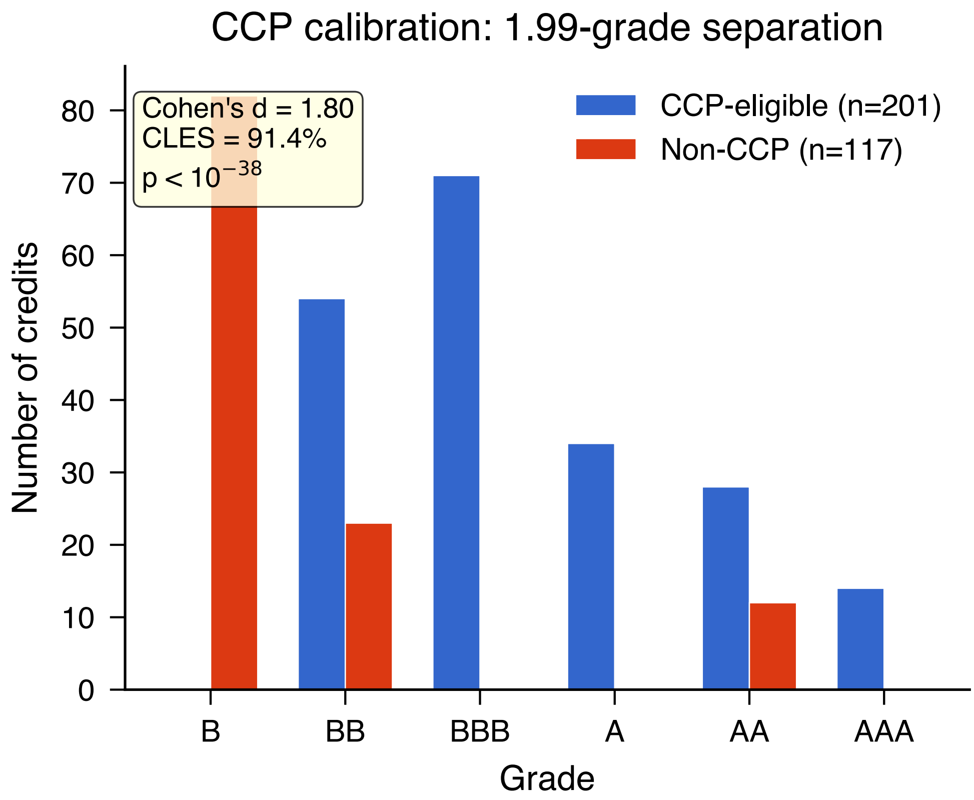

Fig. 5 CCP calibration: quality score distributions for CCP-eligible versus non-CCP credits. Cohen's $d = 1.87$.

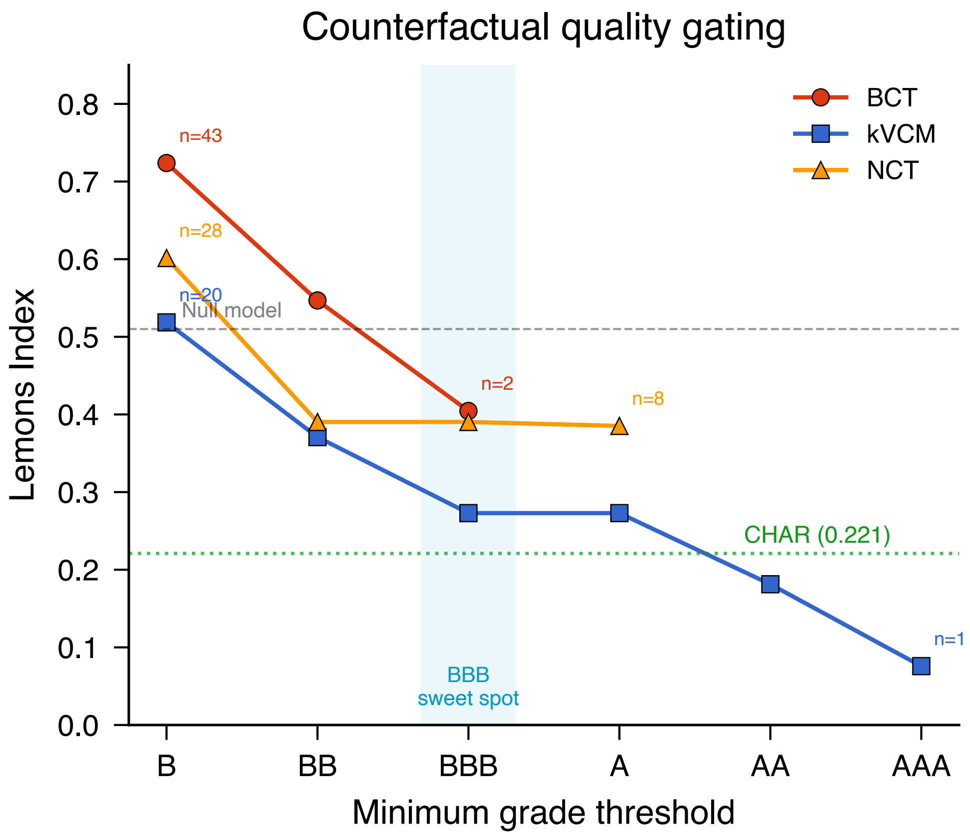

Fig. 6 Counterfactual quality gating: PQD reduction under progressive grade thresholds for four pools.